

VIHARI REDDY ALETI

Data Engineer | Python, SQL, Spark

+1(940)290-3394 | aleti.viharireddy@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

M.S. in Information Technology graduate (3.86/4.0) with hands-on experience building a PostgreSQL-to-Databricks pipeline ingesting 2M+ records across 4 tables and a Kafka-Flink streaming analytics pipeline using Python, SQL, Spark, Databricks, and AWS. Seeking an Entry-Level Data Engineer role, bringing prior professional experience in data analysis and reporting.

EDUCATION

Belhaven University

Master of Science in Information Technology

Jan 2025 – May 2026

Jackson, MS

Jawaharlal Nehru Technological University

Bachelor of Science in Computer Engineering

Aug 2016 – Sep 2020

Hyderabad, India

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, Scala
- **Data Processing & Transformation:** Apache Spark, PySpark, Databricks, dbt, Pandas, Polars
- **Streaming & Orchestration:** Apache Kafka, Apache Airflow
- **Data Platforms & Databases:** Databricks, Snowflake, PostgreSQL, MySQL, DuckDB
- **Cloud & DevOps:** AWS (S3, EMR), Docker, Kubernetes, Git
- **BI & Visualization:** Power BI, Quicksight
- **Machine Learning & GenAI:** Scikit-learn, LLMs, RAG, Prompt Engineering

DATA ENGINEERING PROJECTS

Lakeflow Ingestion Framework | [GitHub](#)

Mar 2026 – Present

Tools: Python, PySpark, Databricks, PostgreSQL, Delta Lake, GitHub Actions

- Built a metadata-driven **PostgreSQL** ingestion framework for full loads and WAL-based CDC into **Databricks** Lakehouse, ingesting 2M+ records across 4 tables.
- Improved pipeline resilience with three-attempt exponential backoff for transient failures and failure classification to prevent retries of non-recoverable errors.
- Validated end-to-end CDC accuracy across **INSERT**, **UPDATE**, and **DELETE**, using Dockerized PostgreSQL and live replication slots.
- Automated CI/CD across dev, QA, Prod workspaces using **GitHub Actions** and Databricks Asset Bundles, enforcing linting, type checking, and test-coverage quality gates.

Agentic Aegis – Self Healing ETL Pipeline | [GitHub](#)

Nov 2025 – Feb 2026

Tools: Pandas, NumPy, DuckDB, Scikit-learn, MySQL, AWS S3, LLM, Docker

- Implemented a Bronze-Silver-Gold ETL pipeline with **Prefect**, **Polars**, and **DuckDB**, integrating 4 source types (CSV, PostgreSQL, MySQL, S3) to produce curated analytics tables.
- Automated data-quality checks across 8 issue categories, using profiling, a weighted quality score, and rule-based remediation to produce auditable cleaned data.
- Enabled governed natural-language analytics by building an **LLM-powered SQL** interface restricted through schema-only context, query validation, and read-only database access.
- Improved platform observability by containerizing services with Docker and exposing Prometheus metrics for pipeline duration, quality scores, and remediation activity.

Real-Time Streaming Analytics Pipeline | [GitHub](#)

Jul 2025 – Oct 2025

Tools: Python, Java, SQL, Kafka, Flink, Spark, Redis, Kubernetes

- Built a **Kafka-Flink** pipeline that refreshes source-level counts, error rates, and latency percentiles every 10 seconds using 1-minute sliding windows.
- Developed 3 stateful anomaly detectors using Z-score, MAD, and EWMA to identify sudden metric spikes and gradual drift in streaming events.

- Designed a dual-engine analytics for a 5-stage conversion funnel using Flink per-event aggregation and **Spark** 30-minute session windows.
- Delivered real-time KPI telemetry through Flink sliding-window updates, Redis hot storage, TimescaleDB historical data, and 2-second WebSocket pushes.

PROFESSIONAL EXPERIENCE

IT World Web
Executive Consultant - Business & Data

Dec 2022 – Dec 2024
Bengaluru, India

- Led UAT for migration of 2M+ customer records to a cloud warehouse; caught 400+ defects pre-launch, zero critical issues in the first 90 days post-cutover.
- Investigated 230 reporting discrepancies in six months, traced 52% to inconsistent business definitions, and drove remediation that reduced repeat exceptions by 34%.
- Cut Weekly Business Review preparation by 4 hours per week by building QuickSight dashboards for operational trend analysis.

Careernet Technologies
Associate Consultant - Business & Data

Apr 2021 – Nov 2022
Bengaluru, India

- Reduced reporting defects by 35% and shortened analytics turnaround by one week per cycle by establishing source-to-report mappings and data validation rules.
- Standardized calculation logic for 10+ enterprise KPIs by reconciling source data with performance reports, resolving conflicting metrics across stakeholder teams.
- Reduced client-requested data corrections from 11 to 6 per month by implementing missing-entry and duplicate entry checks across eight business workflows.

CERTIFICATIONS & TRAININGS

- Analytical SQL for Developers – Oracle
 - Data Engineering on AWS Foundations – AWS
- July 2026
March 2026